

Derivatives & Alternatives, at a glance.

A condensed, exam-ready reference for the whole module — both chapters, plus the **worked numbers that live only in the class deck**. Everything here comes from Novia's own study guide and slides, reduced to the definitions, formulas, directions and traps you are most likely to be tested on.

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§1.1 · DERIVATIVE MARKETS — THE FOUNDATION

Derivative — a **financial contract whose value is derived from the value of some underlying asset**. The underlying may be a share, bond, commodity, currency, interest rate or index — or an **outcome** (weather, carbon emissions, heating and cooling days). Derivatives manage financial risk, exploit investment opportunities, and create **synthetic asset class exposure**.

EXAM WATCH The **primary purpose of derivative markets is the transfer of risk without the need to trade the underlying asset**. Learn that phrase verbatim. Size of the global market: **≈ US\$800 trillion**, against **≈ US\$23 trillion** for every exchange-listed company in the United States combined.

The three main classes

- **Forwards / futures** — priced off spot plus a basis
- **Options** — priced off time, volatility, strike
- **Swaps** — priced by equating the PV of a fixed and a variable stream

Each class has its **own pricing mechanism** — the exam likes matching the class to its price driver.

What moves a derivative's price

- Price of the underlying
- Time to maturity
- Volatility
- Interest rates
- Dividends
- Credit risk & counterparty risk
- Market conditions — liquidity, depth, transaction costs

FRAMING Derivative pricing is a complex process requiring an understanding of the instruments, the mathematical models and market dynamics. It is a **fundamental aspect of financial markets** — institutions, traders and risk managers all use pricing models to make informed decisions.

§1.1.6 · THE FOUR KEY TERMS OF A DERIVATIVE CONTRACT

TERM	WHAT IT FIXES
Underlying	The asset or outcome the contract is built on. Originally all derivatives were based on tangible assets ; some are now based on outcomes . Examples: agricultural products (wheat, rice, soybeans, cotton, butter, milk) · livestock (pigs, cattle) · currencies · interest rates · individual shares and equity indices · bond indices · economic factors such as the inflation rate · natural resources (crude oil, natural gas, gold, silver, timber) · weather-related outcomes such as heating or cooling days.
Size and price	Size = the amount of the underlying to be exchanged. Price = what the underlying will be bought or sold for — the exercise price or strike price . This is not the current or spot price — it is a price good for future delivery.
Expiration date	All derivatives have a finite life ; each contract specifies the date the contract ends. Settlement date is normally two days before final expiration , and settlement instructions are exchanged and confirmed between the counterparties on that date.
Settlement	How the contract is satisfied at expiration. Some require physical delivery (and then the contract specifies delivery locations); others allow or require cash settlement . Contracts on outcomes such as heating or cooling days cannot be physically delivered and must settle in cash. In practice most derivatives contracts settle in cash.

TRAP The **strike price is not the spot price**, and **settlement date ≠ expiration date** — settlement sits two days earlier. Both are favourite one-mark distractors.

Users of derivatives — and what they use them for

The main users are **financial institutions, corporations and institutional investors such as hedge funds**.

PURPOSE	USE
Hedging	Protect against price fluctuations in raw materials, exchange rates or interest rates
Investing	An alternative to investing directly in the assets
Speculating	Take calculated risks based on future price predictions
Arbitrage	Exploit price differences between related assets or markets
Insurance	Protection against adverse events, such as catastrophic events

Other users: **margin traders** — use borrowed funds to increase their trading positions — and **market makers**, who ensure there are always buyers and sellers.

§1.1.9 · OTC VS EXCHANGE-TRADED DERIVATIVES

OTC derivative — a financial contract between two parties that is **not traded on an exchange**. **Customised** to meet each party's needs and **more flexible** than an exchange-traded derivative, but potentially **riskier due to the lack of regulation**.

Exchange-traded derivative — a **standardised** contract such as a future or an option, transacted on an **organised futures exchange**. Standardised, and requiring payment of an **initial deposit or margin**, settled through a **clearing house**.

KEY POINT	OTC DERIVATIVES MARKET	EXCHANGE-TRADED MARKET
Nature of the market	Decentralised	Centralised
Nature of the contract	Tailored / customised	Standardised
Regulation	Not regulated	Highly regulated
Counterparty risk	High	None — the clearing house stands in
Liquidity	Low (instrument dependent)	High
Accessibility	Exclusive — mainly big companies and financial institutions	Accessible to the general public and to big investors
Transparency	Lower	Higher
Credit risk	Increased	Lower
Settlement	At the end of the agreement	Marked to market — settled daily

EXAM WATCH Examples of **OTC**: **swaps** (the most common OTC derivative), **forward contracts**, **forex**, and a **swaption**. Other OTC underlyings: interest rates, commodities, equities, foreign exchange rates, fixed income securities, credit risk. Examples of **ETD**: **futures, options, and options on futures**.

The guide's side-by-side

Exchange-traded contracts have standard terms and trade on an organised derivatives exchange (a futures exchange). They are **tradable financial instruments, marked to market (settled daily)**, with **performance guaranteed by the exchange's clearing house**, and are **highly liquid with an active secondary market** because every contract is standardised.

OTC contracts are **customised transactions created by two parties** through a telephone- and computer-linked dealer network. **Financial institutions often act as market makers** for the more commonly traded instruments, **trades are typically much larger** than in the exchange-traded market, and there is **usually some default risk**.

Benefits — three headings

Price discovery

Futures prices can be thought of as a **forecast of future spot prices** — though in reality they provide only a **little more information than the spot price**, and they do so **efficiently**. A futures price also indicates **what price would be acceptable to avoid uncertainty**.

For options, one characteristic of the underlying is **volatility**; option pricing models let you back out the volatility **implied** by the option price. **Implied volatility is a good measure of general uncertainty in the market — a measure of fear.**

Operational advantages

- **Lower transaction costs** than transacting in the equivalent underlying
- **Greater liquidity** than the underlying market
- **Short positions entered into easily**

Market efficiency

When prices deviate from fundamental value, derivatives offer a **low-cost way to exploit the mispricing** — less capital, lower costs, shorting possible. Investors are **far more willing to trade if they can manage their risks**, raising participant numbers and therefore **market liquidity**.

Criticisms — two headings

Speculation and gambling

For hedging to work there **must be speculators on the other side of the trade** — and the more speculators the market attracts, the **cheaper it becomes to hedge**. Yet speculators are perceived poorly: short-term traders thought to engage in price manipulation and trading at extreme prices. Profit from short-term trading is **taxed more heavily** than long-term profit — a way of “punishing” the activity. Many view derivatives trading as **legalised gambling**, but gambling **benefits only a limited number of participants and does not help society as a whole**, whereas the derivatives market brings **extensive benefits to the financial services industry**.

Destabilisation and systemic risk

Opponents argue the operational benefits produce **excessive speculative trading** and therefore instability. Speculators use **large amounts of leverage**, exposing themselves and their creditors to high risk; defaults by speculators can lead to defaults by their creditors, and these **chain-reaction events can be systemic**.

The second criticism is **complexity**: when the models underpinning derivative pricing break down under sometimes irrational market behaviour, **the model builders are blamed** for failing to capture financial market reality.

QUOTABLE Warren Buffett called derivatives “**financial weapons of mass destruction**”. The deck’s framing: a leveraged instrument can **not only reduce risk but create risk** through enhanced uncovered exposure. The three legitimate purposes: **hedge** (contain upside and downside) · **insure** (retain the upside, limit the downside) · **transform** assets and liabilities to match future expectations.

§1.1.7 · CONCEPTS AND TERMINOLOGY (1 OF 2)

Arbitrage — a trading strategy **requiring no cash** where there is some probability of making a profit **without incurring the risk of a loss**. The **simultaneous purchase and sale of identical or equivalent financial assets** to benefit from a discrepancy in their price relationship. A **low-risk** strategy linked to **hedging (risk transference)** and **speculation (risk acceptance)**.

TYPE OF ARBITRAGE	WHAT IT TRADES
True arbitrage	Two assets with an equal fair value trading at different prices .
Risk arbitrage	An asset currently priced at a value that will soon change .
Convertible arbitrage	The difference between a bond's conversion price and the current price of the underlying company's shares.
Triangular arbitrage	Three or more currencies traded simultaneously.

Speculation — buying and selling assets to **profit from price fluctuations over the short term**, as opposed to buying for long-term gains (buy and hold) or income (dividends). Associated with **day trading**, it relies on **market timing** and implies a **riskier proposition than investing**. It can involve any tradable asset — shares, bonds, commodity futures, currencies, cryptocurrency, fine art, collectibles, real estate and financial derivatives. Speculators **provide liquidity and help narrow the bid-ask spread**.

Hedging — offsetting the price risk inherent in a spot market position by taking an **equal but opposite position in the futures market**. Risk is **neutralised and transferred** to someone for whom holding that risk would **reduce an existing but opposite exposure**. A counterparty **without** a pre-existing exposure — a **speculator** — accepts the risk to benefit from an anticipated price movement.

Strategies: **options hedging** (e.g. buy a put to protect against a share price decline) · **currency hedging** (buy or sell a foreign currency to offset exchange rate effects) · **direct hedging** (open a position in the opposite direction of an existing trade, e.g. a sell position in the same pair you are long).

Hedging is used by short- to mid-term traders and investors — not typically as part of a long-term strategy, because short-term price fluctuations have little impact on buy-and-hold investors.

Marking to market — the mark-to-market process, or **margining**, ensures that all counterparties **meet their obligations** under the futures contract. The **initial margin determines the level of gearing** obtained from entering into a futures contract.

Volatility — the **degree of fluctuation (rate of change)** in the price of a security, expressed mathematically as **variance or standard deviation, and beta**. The greater the volatility, the wider the fluctuations between successive prices over a given period. **The volatility of security returns is the measurement used to define risk**. High volatility means considerable risk and an increased probability of either a loss or a gain in a short period. **Futures markets are traditionally more volatile than spot markets.**

Short selling — a strategy where a security is **borrowed and sold** with a commitment to **purchase and replace it later**. A complex, high-risk strategy of selling a security not owned, used to capitalise on an expected decline in price, which can lead to substantial losses if the opposite occurs. There are **many restrictions on selling short**.

Short selling applies to the spot or cash market. "Going short" or "shorting an asset" applies to the derivatives market.

Market maker — a **JSE member** — for example the investment banks **ABSA, RMB, Investec and Standard Bank** — that **always quotes bid and ask prices** for selected instruments, thereby making or maintaining a market in them. **Market makers must cover or hedge their exposure in the futures market with equivalent transactions in the spot market** — they buy or sell the client's full long or short exposure in the spot market.

EXAM WATCH The two terminology traps that recur: **short selling (spot) versus going short (derivatives)**, and **arbitrage carries no risk of loss and requires no cash** — if an answer option describes a strategy with downside risk, it is not arbitrage.

§1.2.1 · FORWARD COMMITMENTS VS CONTINGENT CLAIMS

Forward commitments

Contracts in which the two parties agree to engage in a transaction **at a later date at a price established at the start**. They create an **obligation for both parties**. Two major classifications: **exchange-traded** contracts (futures) and **over-the-counter** contracts (forwards and swaps).

Forwards

Futures

Swaps

Contingent claims

Contracts in which **the payoffs occur if a specific event happens**. They create the **right — but not the obligation** — for one party to buy or sell an underlying asset at a fixed price over a specific period.

Options

Asset-backed securities

Credit derivatives

EXAM WATCH **Forward commitment = obligation. Contingent claim = right.** That difference changes the **payoff profile**, which is why the two trade differently. And **a swap is equivalent to a series of forward contracts**.

INSTRUMENT	CHARACTERISTIC
Forward contracts	Customised agreements
Futures contracts	Standardised agreements
Swaps	Equivalent to a series of forward contracts
Options	Give the buyer the right to buy or sell the underlying at a specific price within a certain period
Call options	Give the buyer the right to buy the underlying
Put options	Give the buyer the right to sell the underlying

Forwards vs futures — the six differences

	FORWARDS	FUTURES
Regulation	Not formally regulated	Regulated by the JSE / SAFEX
Trading	Over the counter	On an exchange
Contract terms	Customisable	Standardised
Delivery dates	Usually one specified delivery date	A range of delivery dates
Settlement	Settles at the end of the agreement	Settled daily until the end of the contract
Counterparty risk	More counterparty risk	Less — the clearing house guarantee

Forward contract — an **over-the-counter derivative contract in which two parties agree that one party, the buyer, will purchase an underlying asset from the other party, the seller, at a later date at a fixed price they agree on when the contract is signed.** The parties also agree the **identity of the underlying**, the **number of units** to be delivered and **where delivery will occur.** The contract is **customised**, and **each party is subject to the possibility that the other will default.**

The four types of forward

TYPE	DEFINITION
Equity forwards	Binding agreements to buy or sell a specific stock or the return on a share index at a fixed entry price at a fixed expiry date.
Fixed-income forwards	An agreement to transact in a fixed-income security at a preset price at a future forward date. Value = bond price – PV of coupon payments – PV of the price at expiration.
Interest rate forwards also called FRAs	Forecasts that estimate future interest rates, used to plan for economic shifts. Calculated from the spot rate — the rate starting today — adjusted for the cost of carry.
Currency forwards forward exchange contracts, FECs	A binding agreement that locks in an exchange rate for a future currency transaction. Privately negotiated and traded OTC in the foreign exchange market.

Settling a forward at expiration

Delivery — the long party pays the agreed-upon price; the short party delivers the underlying asset.

Cash settlement — the net position is settled:

the long receives money when Spot > Future

the short receives money when Future > Spot

The parties

The party that agrees to **buy** the asset has a **long forward position**. The party that agrees to **sell** has a **short forward position**.

Dealers are financial institutions quoting bid (buy) and ask (sell) prices, earning the spread. **End users** are companies or institutions **managing risk** — **locking in a future price**.

CREDIT RISK IN FORWARD CONTRACTS There is credit risk because **the counterparty may not deliver the asset at the time of delivery.** And because a forward is **not exchange traded**, a **buyer or seller cannot lock in gains or losses on the contract's value before the agreed settlement date** — **the contract must be held to maturity.** Given this, **the contract's value serves as the quantification of credit risk** for the counterparties.

Futures contract — a **standardised derivative contract created and traded on a futures exchange** in which two parties agree that one party, the buyer, will purchase an underlying asset from the other party, the seller, at a later date and at a price agreed on by the two parties when the contract is initiated, **and in which there is a daily settling of gains and losses and a credit guarantee by the futures exchange through its clearinghouse.**

EXAM WATCH The **most important distinctive characteristic of a futures contract** is the **daily settlement of gains and losses and the associated credit guarantee** provided by the exchange through its clearinghouse. Learn the full definition above — the three elements in bold are where the marks are.

What standardisation buys you

- Futures are **specialised versions of forward contracts** that have been **standardised and trade on a futures exchange**
- Standardised contracts + an organised market with rules and regulations + a **central clearing facility** deliver **liquidity and protection against loss by default**
- A futures contract requires a **futures exchange** — a **legally recognised entity** providing a market for trading these contracts
- Futures exchanges are **highly regulated at the national level in all countries**, and **specify that only certain contracts are authorised for trading**
- Those contracts have specific **underlying assets, times to expiration, delivery and settlement conditions and quantities**
- The exchange offers a **physical location and/or electronic system**, plus **liquidity provided by authorised market makers**
- Futures are **guaranteed by the clearinghouse — no counterparty risk**; gains and losses are **settled daily (marked to market)**; and **margin is required as collateral to cover losses**

CONTRAST WITH FORWARDS In a forward contract the parties **customise everything** — the underlying, time to expiration, delivery and settlement conditions and quantity — **according to whatever terms they agree**. These contracts are **not traded on an exchange**, and although OTC regulation is increasing, they are **not subject to the high degree of regulation that applies to securities and futures markets.**

Like forward markets, futures markets can be used for **hedging or speculation** — for example a **jewellery manufacturer can buy gold futures**, hedging the price it will have to pay for one of its key inputs.

Mark to market — the daily settlement

The agreed-upon price is the **futures price**. Identical contracts trade on an ongoing basis at different prices, reflecting the passage of time and the arrival of new information — so as the futures price changes, **the parties make or lose money**. Rising prices **benefit the long and hurt the short**; falling prices do the reverse.

At the end of each day the clearinghouse **determines an average of the final futures trades of the day** and designates that price the **settlement price**. All contracts are then **marked to the settlement price**.

Worked example — the guide's own

The long purchases the contract during the day at a futures price of **R120**. The settlement price at the end of the day is **R122**.

The long's account is **marked for a gain of R2** — that amount is **credited to his account, with the money coming from the account of the short, who has lost R2**. If the futures price decreases instead, the long loses money and is charged with that loss, and the money is **transferred to the account of the short**.

Futures pricing — the class deck's formula

$$F = S (1 + r)^t$$

Deck worked example: spot **R120**, rate **8%**, one year → $F = 120(1.08)^1 = \mathbf{R129.60}$. This is the **cost-of-carry** relationship — the futures price is the spot price carried forward at the financing rate, which is why the guide describes a futures price as “**the spot price along with a basis amount**”.

Payoff shape

A futures contract is an **obligation**, so it exhibits a **symmetrical payoff** — gain and loss run in both directions without limit. Options, being **rights**, are **asymmetrical**: the buyer's loss is capped at the premium.

Hedge profiles

Long futures locks in a specific purchase price.

Short futures locks in a specific selling price.

Both **eliminate any downside (loss)** — but also **forfeit any upside (gain)**.

EXAM WATCH Mark to market is the mechanism that makes futures **credit-safe**: the gain of one party is funded **daily** out of the account of the other, so **no large unsettled exposure accumulates** — which is precisely what a forward contract does allow.

§1.3 · OPTIONS — DEFINITION AND CONCEPTS

Option — a derivative contract in which one party, the **buyer**, pays a sum of money to the other party, the **seller or writer**, and receives the **right to either buy or sell an underlying asset at a fixed price**, either on a specific expiration date or at any time prior to it. **The purchase price of the option is the premium**; sellers of options **receive premium income**.

Call option

Grants the **holder (long position)** the opportunity to **buy** the underlying security **at a price below the current market price**, provided the market price **exceeds the call strike** before or at expiration.

Put option

Grants the **holder (long position)** the opportunity to **sell** the underlying security **at a price above the current market price**, provided the **put strike exceeds the market price** before or at expiration.

In both cases the **option seller (short position)** receives a payment — the **premium** — **compelling performance at the discretion of the holder**. If the holder exercises, the writer must **make delivery (call)** or **take delivery (put)**. Combinations of options lead to different exposures; the incentive to contract as long or short centres on **perceptions regarding market direction and expectations**.

The six options concepts

TERM	DEFINITION
European-style option	An option that can be exercised and settled only on its expiration date .
American-style option	An option that can be exercised and settled on or before its expiration date — the most common type .
Moneyness	The degree to which an option is in the money and likely to be exercised , thereby having a payoff.
Payoff	The difference between the spot and strike prices when exercised.
Breakeven price	The spot price at which the holder breaks even — zero profit — after accounting for the cost of obtaining the option.
Profit or loss	The payoff minus the cost of obtaining the option (the premium) .

§1.3.2 · MONEYNES, INTRINSIC VALUE AND TIME VALUE

	CALL	PUT
In the money (ITM) has intrinsic value — you could exercise immediately for a profit opportunity	Strike is below the current market price	Strike is above the current market price
At the money (ATM) no moneyness	Strike price exactly equals the current price of the underlying — you would neither make nor lose money by exercising	
Out of the money (OTM) no intrinsic value, only time value	Strike is higher than the current market	Strike is lower than the market

Moneyness tells the holder whether exercising will lead to a profit, looking at the value of the option **if you were to exercise it right away**. Guide example: a call to buy XYZ at **R40** while XYZ trades at **R45** is **in the money** — you could exercise to buy at a discount. **The higher the market price relative to a call's strike, the more in the money it is**. A call to buy at **R50** while the shares trade at **R45** is **out of the money** — exercising would mean paying more than the market.

Intrinsic value and time value

The option premium breaks into two components — **intrinsic value** and **speculative or time value (extrinsic value)**.

Market Price – Strike Price
= intrinsic value of a **call**

Strike Price – Market Price
= intrinsic value of a **put**

In both cases **multiply by the number of shares in the contract — typically one hundred**. Intrinsic value represents the profit the holder would book if they exercised, took delivery and sold the asset in the current marketplace.

TIME VALUE Time value is harder to quantify. The more time before expiry, the more opportunity for the option's moneyness to increase — so **time value is higher the longer to expiration, and reduces over time until, at expiration, it reaches R0**. American-type options can be exercised at any point up to that date.

§1.3 · OPTION PAYOFFS — THE FORMULAS AND TWO WORKED EXAMPLES

Payoff to the call holder

$$\text{Call payoff} = \max(0, S_T - X)$$

0 if $S_T < X$

$(S_T - X)$ if $S_T > X$

$$\text{Profit} = \text{Payoff} - \text{Premium}$$

Payoff to the put holder

$$\text{Put payoff} = \max(0, X - S_T)$$

0 if $S_T > X$

$(X - S_T)$ if $S_T < X$

$$\text{Profit} = \text{Payoff} - \text{Premium}$$

Worked example 1 — buying a call (guide §1.2.3)

One call option contract on ABC covers **100 shares**. Strike **R25**, premium paid **R150**. On expiration date ABC shares are selling for **R35**.

Cost of exercising — $100 \times \text{R}25$	R2,500
Sale proceeds — $100 \times \text{R}35$	R3,500
Gross profit on the option — $\text{R}3,500 - \text{R}2,500$	R1,000
Less the premium paid	–R150
Net profit, excluding transaction costs	R850

A very good return on a R150 investment. **The buyer's maximum loss is always limited to the premium** and the upside is **unlimited** — the guide calls this asymmetry “**one of the most attractive features of buying options**”.

Worked example 2 — breakeven on a call (class deck)

An IBM share trades at **R100**. An April call has a premium of **R5** and an exercise price of **R100**. Ignoring commissions:

$$\text{Breakeven} = \text{Strike} + \text{Premium} = \text{R}100 + \text{R}5 = \text{R}105$$

The share price must rise **above R105** for the option holder to earn a profit. Between R100 and R105 the option is in the money but the holder is still down on the premium; below R100 it expires worthless and the loss is the R5 premium.

EXAM WATCH A payoff question and a profit question are **not the same question**. **Payoff ignores the premium; profit subtracts it**. Read which one is being asked.

Worked example 3 — the protective put

An investor owns **1,000 Sasol shares** priced at **R28**. He is concerned about a possible decline over the next two months and wants protection. He buys **ten July R27.50 put option contracts**, giving him the right to sell 1,000 shares at R27.50. The put premium is **R1 — R100 cost per put**.

Total cost of the hedging strategy — $10 \times R100$	R1,000
Proceeds if exercised — $1,000 \times R27.50$	R27,500
Amount realised once the cost of the options is taken into account	R26,500

The strategy **guarantees the shares can be sold for R27.50 during the life of the option**. If Sasol stays **above R27.50**, the options **expire worthless** and the R1,000 was the cost of the insurance.

The four principles of hedging with options

- Hedging means taking a position in some market that will **limit or protect the downside** in another position
- Calls and puts allow **a great deal of flexibility** in creating a hedge
- **Protective puts establish a downside floor; selling a call against an existing position generates income while limiting upside potential**
- Hedging with options **incurs a cost** — understand the risks and rewards of each position

Writing calls — covered vs naked

TYPE	POSITION AND RISK
Covered call	The seller actually owns the underlying stock . Selling calls on stock you own produces additional income and offsets expected declines in the price. The seller is “covered” against a loss — if the buyer exercises, he delivers shares he already owns. His profit is limited to the stock's rise to the strike price , but he is protected against any actual loss .
Naked call	The seller does not own the underlying. Very risky — there is no limit to how high a share price can go , and the seller is not covered. On exercise the naked seller must buy the stock at the current market price to deliver; the difference between market price and strike is his loss . Most option sellers charge a high fee to compensate .

THE WRITER'S SIDE GENERALLY The call seller's downside is **potentially unlimited** — as spot exceeds the strike the writer's loss **equals the buyer's profit**. If the market price never goes higher than the strike, **the option expires worthless and the seller keeps the premium**. Deck example: a call on RBC with a **R10 premium** and a **R100 strike** — the buyer loses R10 if RBC does not rise past R100; **the writer is in the money as long as the share stays below R110**. For the equivalent put, the **writer is in the money while the share stays above R90**.

§1.3.3 · THE SIX FACTORS THAT DETERMINE AN OPTION'S PRICE

This table is the highest-yield page in the chapter. The exam tests **direction**: does the factor push the call up or down, and the put up or down.

FACTOR	CALL	PUT	WHY
↑ Price of the underlying	↑	↓	A higher underlying means greater intrinsic value for a call. For puts the relationship is reversed.
↑ Exercise (strike) price	↓	↑	A higher exercise price is worse for a right to buy and better for a right to sell .
↑ Risk-free rate	↑	↓	Reflects borrowing costs . An increase raises call values and lowers put values.
↑ Volatility of the underlying	↑	↑	Both rise. Volatility is what makes options valuable — with no volatility, options would always equal their intrinsic value and time value would be zero .
↑ Time to expiration	↑	↑	Both rise. Volatility is expressed per unit of time , so more time effectively increases expected volatility and gives higher chances of the option moving in the money .
↑ Costs and benefits of holding the asset dividends, interest, convenience yield	↓	↑	A cash benefit reduces the value of the asset itself — a dividend or coupon lowers the asset's value — and a fall in the underlying decreases call values and increases put values .

EXAM WATCH Only two factors move calls and puts in the same direction: **volatility and time to expiration**. Every other factor moves them in opposite directions. If you remember nothing else on this page, remember that.

Option pricing models

The **Black-Scholes model** is **the most widely known model** used to determine the reasonable value of an option. Other models include the **binomial** and **trinomial** models.

Option prices can be **more volatile than share prices**, which is what attracts traders — but some option strategies carry minimal risk and can even **enhance stock returns**.

Deck mechanics — the \$1.55 pair

You hold a **call** struck at **\$90**, market **\$91** → **exercise**: why pay more when you have something that lets you buy cheaper?

You hold a **put** struck at **\$91**, market **\$90** → **exercise**: why get less when you have something that lets you sell higher? In each case the **only loss possible is the \$1.55 premium already paid**.

§1.4 · SWAPS — DEFINITION AND INTEREST RATE SWAPS

Swap — an **over-the-counter derivative contract** in which two parties agree to exchange a **series of cash flows** whereby one party pays a **variable series determined by an underlying asset or rate** and the other pays either **(1) a variable series determined by a different underlying asset or rate**, or **(2) a fixed series**.

Key characteristics

Cash flows — one is fixed, the other variable, based on an index price, interest rate or exchange rate. **Principal** — the principal amount **usually does not change hands** during or at the end of the swap. **Trading** — OTC or on security-based **Swap Execution Facilities (SEFs)**. **Types** — the most common are **interest rate and currency swaps**.

Where a swap sits

A swap is **more like a forward contract — a series of forward contracts — than a futures contract**: it is an OTC contract, **privately negotiated and subject to default**. Other terms — the identity of the underlying, payment dates and payment procedure — are **negotiated between the parties and written into the contract**.

Interest rate swap — a forward contract in which **one stream of future interest payments is exchanged for another based on a specified principal amount**. Usually a **fixed rate payment is exchanged for a floating rate payment, or vice versa**, to reduce or increase exposure to interest rate fluctuations, or to obtain a **marginally lower interest rate** than would otherwise be possible. Also called **plain vanilla swaps**, being the original and simplest such instruments.

The three types of interest rate swap

TYPE	STRUCTURE
Fixed-to-floating	A company that can issue a bond at an attractive fixed rate but wants better cash flow from a floating rate enters a swap in which it receives a fixed rate and pays a floating rate . The swap is structured to match the maturity and cash flow of the fixed-rate bond , and the two payment streams are netted . The company receives SOFR plus or minus a spread reflecting both market interest rate conditions and its credit rating.
Floating-to-fixed	A company with no access to a fixed-rate loan borrows at a floating rate and enters a swap to acquire a fixed rate . The loan's floating-rate tenor, reset and payment dates are mirrored on the swap and netted , and the fixed-rate leg of the swap becomes the company's borrowing rate .
Float-to-float known as a basis swap	Changes the type or tenor of the floating rate index paid — for example from three-month SOFR to six-month SOFR because the rate is more attractive or matches other payment flows, or to a different index such as the federal funds rate, commercial paper or the Treasury bill rate .

EXAM WATCH The Secured Overnight Financing Rate (SOFR) replaced LIBOR as the benchmark index for interest rate swaps. And a **received-fixed swap should be treated as buying a fixed-rate bond and issuing a floating-rate bond**.

The five motivations for a swap

Hedging against interest rate risk — convert fixed-rate debt into floating, or vice versa · **customising cash flow** to align with financial goals · **accessing new markets** and rates otherwise unavailable · **optimising debt structure** · **converting foreign currency payments** into local currency for bonds issued abroad.

As in most financial transactions, a **swap dealer sits between the two parties and takes a commission on the trade**. Deck example of a plain vanilla swap: **Company A pays LIBOR + 1.1% while Bank B pays a fixed 4.7%**.

Currency swaps

Guide example: **Party A pays Party B \$10 million in return for €9.8 million**. On each settlement date **Party A, who received euros, pays 6% interest in euros on €9.8 million**, and **Party B pays 5% interest on the \$10 million received**. There is no **netting of payments** — they are in different currencies.

Motivation: a company wants to establish operations in a foreign country and finance the costs in that currency, but it is **more expensive to issue debt in the foreign country**. It issues debt in its local currency and enters a currency swap to exchange for the foreign currency; **the counterparty faces a similar situation in reverse**, and an intermediary swap dealer facilitates.

Currency swaps also let a company **transform a foreign currency bond into a domestic obligation** — a receive-fixed foreign currency, pay fixed or floating domestic currency swap — and likewise transform a foreign investment into a domestic one.

The four possible types: pay fixed on foreign / receive fixed on local · pay floating on foreign / receive fixed on local · pay fixed on foreign / receive floating on local · pay floating on foreign / receive floating on local.

Swaption — an option on a swap: an option to **enter into a swap at a fixed rate**. The most widely used is the **plain vanilla interest rate swaption**. It lets the holder **establish a fixed rate on the underlying swap in advance** and then choose either to enter the swap at that rate or **let the swaption expire** and enter at the prevailing market rate.

Payer swaption — an option on a swap to **pay fixed and receive floating**; the buyer **hopes the fixed rate goes up** before expiry. **Receiver swaption** — an option to **receive fixed and pay floating**; the buyer **hopes the fixed rate goes down**.

Two motivations: **to change the pattern of payments on an upcoming fixed-income obligation** — a payer swaption protects against a rate increase, and if the market rate at expiry is below the contract rate the buyer **walks away and pays the lower market rate** — and **to terminate a swap position**, the swaption acting as an insurance policy if the original swap goes wrong.

As with all options, **the buyer pays a premium for this asymmetric payoff** — the right to walk away.

CREDIT DEFAULT SWAP (CDS) A credit derivative contract enabling investors to swap credit risk on a company, country or other entity with another counterparty. **The most common type of OTC credit derivative**, often used to **transfer credit exposure on fixed income products in order to hedge risk**. If a lender fears a borrower will default, it **buys a CDS from another investor who agrees to reimburse it if the borrower defaults**. Most CDS require an ongoing premium payment to maintain the contract — like an insurance policy.

EXAM WATCH South Africa's derivatives market is **the only one in Sub-Saharan Africa**. Like other emerging derivatives markets, its development **stemmed primarily from the need to “self-insure” against volatile capital flows** and to manage the financial risk associated with the **high volatility of asset prices** — not simply from a wish to deepen the market.

The four key agencies

AGENCY	ROLE
FSCA Financial Sector Conduct Authority	Established by the Financial Sector Regulation Act 9 of 2017 to become a dedicated market conduct authority , replacing the Financial Services Board (FSB) on 1 April 2018 . Supervises the activities of non-bank financial institutions and other financial services.
The Capital Markets Department of the FSCA	Responsible for the supervision of licensed exchanges, central securities depositories and clearing houses .
The JSE and the BESA	Licensed exchanges trading in derivative instruments, supervised by the FSCA . When the JSE took over the business operations of SAFEX in 2001 , two divisions were created within the JSE: the Equity Derivatives Division and the Agricultural Products Division .
The SAFEX Clearing Company	Under the direction of the JSE and licensed to clear transactions in derivatives traded on the JSE .

Why a derivatives market matters — and what it costs

Derivatives markets **facilitate the management of financial risk exposure** because they let investors **unbundle and transfer financial risk**. In principle they contribute to a **more efficient allocation of capital and cross-border capital flow**, create **more opportunities for portfolio diversification**, and facilitate **risk transfer, price discovery and more public information**. South Africa's market **has grown rapidly**, supporting capital inflows and helping participants price, unbundle and transfer risk — but **the risks associated with misuse have also increased**. There are **tight regulations on asset allocations by insurance and pension funds** to prevent excessive risk taking, and **such regulations can constrain the potential benefits** the local derivatives market could bring. **Misuse can lead to a financial crisis, accelerate capital outflows and amplify volatility — but the advantages should not be discounted.**

§1.5 · WHO TRADES WHAT, AND WHERE

Trading in South Africa's derivatives takes place **both in the over-the-counter market and on the established stock exchanges.**

INSTRUMENT	VENUE
Warrants, equity futures and options, agricultural commodity futures and options	The JSE
Fixed-income derivatives — bond futures, FRAs, vanilla swaps and standard bond options	The BESA
Interest rate futures and options	Both exchanges
Currency futures and fixed-income derivatives such as interest rate futures	Concentrated in the OTC market
Exchange-based interest rate futures and options — commenced 2003	BESA, through Intersec , a fixed-income derivative platform
Exchange-based currency futures — commenced 2007	The JSE currency futures exchange platform Yield-X

REGULATORY NUANCE — WORTH A MARK ON ITS OWN Unlike the BESA, the JSE regulates exchange trading in derivative instruments. The BESA instead provides the platform and regulates a range of derivative instruments on both the exchange and the OTC market for fixed-income derivatives, through Intersec.

Transparency

Market activities in the organised exchanges are more transparent than in the OTC markets. Derivative instruments traded on the floor of the JSE and the BESA are **standardised and transparent**. With the exception of OTC arrangements on Intersec, derivative instruments traded on OTC markets are privately negotiated and bilaterally arranged.

EXAM WATCH · THE DATES SAFEX absorbed by the JSE — 2001. Interest rate futures on BESA via Intersec — 2003. Currency futures on Yield-X — 2007. FSCA replaces the FSB — 1 April 2018, under the Financial Sector Regulation Act 9 of 2017.

Alternative investments — collectively, the many asset classes that **fall outside the traditional definitions of stocks and bonds**. They differ from traditional investments (publicly traded stocks, bonds, cash) **both in the types of assets and securities included and in the structure of the investment vehicles** in which those assets are held.

Hedge funds

Private equity

Real estate

Commodities

Infrastructure

Other — primarily collectibles

Five distinguishing characteristics

- **Lower liquidity** — due to their **lack of standard markets** and limited activity on both sides of the deal
- **Less regulation** — but **unique legal and tax considerations**
- **Lower transparency** — certain alternatives **lack an efficient market mechanism**, subjecting valuation to speculation; risk increases in the **absence of ready valuation information**
- **Higher fees** — costs of purchase and sale may be relatively high
- **Limited and potentially problematic historical risk and return data** — higher-than-normal returns may be subject to a **variety of biases**, and **the volatility of returns tends to be underestimated**

The three key reasons for adding them to a portfolio

REASON	RATIONALE
Reduced risk through diversification	The primary goal. One distinguishing feature of most alternatives is their lack of correlation with public equities and public fixed income . A portfolio containing a variety of alternative assets may offer reduced risk without a proportionate reduction in expected return .
Enhanced return through alpha	Acquiring assets that offer reasonable expectations of alpha — superior risk-adjusted returns . Hedge funds and private equity have a track record of offering such opportunities.
Avoiding obsolescence	The asset classes viewed as appropriate for institutional investing have changed dramatically over time and will keep changing. Institutions that are the last adopters will find prices have already adjusted and the greatest opportunities missed.

MANAGEMENT STYLE Portfolio managers invest either **passively** — index or asset-class coverage, which some real estate, commodity and infrastructure portfolios use — or **actively**. **Alternative investments are generally actively managed**. Beware the **naive attraction** of expected returns without consideration of their **non-typical risks**: low liquidity, limited redemption availability, limited transparency and the challenge of manager diversification.

§2.1.3 · THE CATEGORIES OF ALTERNATIVE INVESTMENTS

CATEGORY	WHAT IT COVERS
Hedge funds	May use leverage, hold long and short positions, use derivatives and invest in illiquid assets . Managers use a great many different strategies to generate investment gains. They do not necessarily hedge risk as the name might imply.
Private equity funds	Invest in the equity of companies that are not publicly traded , or in publicly traded firms the fund intends to take private . Leveraged buyout (LBO) funds use borrowed money to purchase equity in established companies and comprise the majority of private equity investment funds. A much smaller portion, venture capital funds , invest in or finance young unproven companies at various early stages. Investing in the securities of financially distressed companies is also treated as private equity here, though hedge funds may hold these too.
Real estate	Residential or commercial properties as well as real-estate-backed debt . Held in a variety of structures: full or leveraged ownership of individual properties, individual real-estate-backed loans, private and publicly traded securities backed by pools of properties or mortgages , and limited partnerships .
Commodities	Exposure through physical commodities, commodities derivatives, or the equity of commodity-producing firms . Some funds seek the returns of commodity indices , often by holding derivatives contracts expected to track a specific index.
Infrastructure	Long-lived assets that provide public services . Economic infrastructure — roads, airports, utility grids. Social infrastructure — schools, hospitals. Often financed and constructed by governmental entities, but more recently undertaken by private-public partnerships , each holding a significant stake.
Other	Tangible collectible assets — fine wines, stamps, automobiles, antique furniture and art — as well as patents, an intangible asset .

EXAM WATCH Three category facts the exam likes: **LBO funds are the majority of private equity; infrastructure splits into economic and social; and “other” includes patents** — an intangible sitting among the tangible collectibles.

Each of these has **unique characteristics requiring a different approach by the analyst** — different strategies, fee structures, due diligence, and issues in **valuing and calculating returns**.

The four risks of alternative asset funds

RISK	DESCRIPTION
Market risk	Related to the overall stock market trend — the funds may experience sharp short-term fluctuations in value .
Geographical concentration	The rise or fall in value of investments concentrated in a specific geographical region may deviate considerably from the overall stock market trend .
Liquidity risk	Associated with exceptional market situations — if the securities the funds have invested in cannot be converted into cash on the desired schedule or for a reasonable price .
Exchange-rate risk	The higher the number of investments in foreign currencies and the more those currencies fluctuate, the higher the exchange risk. Weakening of the currencies of the fund's investments has a negative impact on the fund's value; strengthening has a positive impact .

The value of alternative asset funds is calculated **daily**, based on the **current market value of the securities the funds own**. Fund values **may fluctuate and deviate significantly from the overall market trend**.

The partnership structure

A **partnership structure is common for many alternative investments** — hedge funds, infrastructure funds and private equity funds. The **fund manager is the general partner (GP)** and **investors are limited partners (LPs)**.

General partner (GP)

The fund manager. **Runs the business and theoretically bears unlimited liability** for anything that might go wrong.

Limited partners (LPs)

The investors. **Own a fractional interest in the partnership** based on their investment amount and the terms of the partnership documentation.

Why the structure matters

Limited partnerships are **restricted to investors expected to understand and to be able to assume the risks**. Because they are **not offered to the general public they are less regulated** than public offerings. They are **frequently located in tax-efficient locations**, which benefits both the GP and the LPs, and funds set up as private investment partnerships **typically have a limit on the number of LPs**.

§2.1.7 · THE FEE STRUCTURE

Four terms you must be able to define and distinguish. They recur in the hedge fund chapter and carry the calculation marks.

TERM	DEFINITION
Management fee sometimes called the “base fee”	Based on assets under management . Earned regardless of investment performance.
Incentive fee or performance fee	Based on realised profits .
Hurdle rate	A specified return the fund must achieve before the incentive fee is earned. Set either as a percentage (e.g. 4%) or a rate plus a premium (e.g. LIBOR + 2%). Hard hurdle rate — incentive fees are earned only on returns in excess of the benchmark. Soft hurdle rate — incentive fees are paid on all profits, but only if the hurdle rate is met.
High-water mark	The highest cumulative return — the highest value, net of fees — of the fund investment by the individual LP , previously recorded. The incentive fee is not paid on gains that merely offset prior losses : incentive fees are only paid to the extent the current value is above the highest value after fees previously recorded . Not all LPs will have the same high-water mark , because it depends on the timing of their individual investment.

EXAM WATCH The examinable purpose of the high-water mark is one sentence: **it protects clients from paying twice for the same performance**. And the hard/soft distinction: **hard = only the excess is fee-bearing; soft = everything is fee-bearing once the hurdle is cleared**.

How the pieces fit together

Funds are generally structured with a **management fee based on assets under management** plus an **incentive fee based on realised profits**. Sometimes the structure specifies that the **incentive fee is earned only after the fund achieves a specified return — the hurdle rate**. Fee calculations also take into account the **high-water mark**. Note too that the incentive fee may be calculated **on profits net of management fees**, or **on profits before management fees** — that is, the incentive fee calculated **independent of management fees**. **Read which basis the question specifies**.

§2.2 · HEDGE FUNDS — WHAT THEY ARE

Hedge fund — an **investment fund that pools capital from individual or institutional investors**, administered by a **professional investment management firm**, investing across assets ranging from vanilla to complex. A contemporary hedge fund is an **aggressively managed portfolio of investments across asset classes and regions that is leveraged, takes long and short positions, and/or uses derivatives**, with a goal of **generating high returns — either in an absolute sense or over a specified market benchmark — and few, if any, investment restrictions**.

Structural characteristics

- Set up as a **private investment partnership open to a limited number of investors** willing and able to make a **large initial investment**
- **Restrictions on redemptions** — investors may be required to keep money in the fund for a minimum period, the **lockup period**, before they may withdraw or redeem shares
- Investors may be required to **give notice of intent to redeem — typically 30 to 90 days — and may be charged a fee to redeem** shares
- Managers typically invest a **significant amount of personal capital — in some cases in excess of 50% of the total assets in the fund** — aligning their interests with those of investors
- Investors **share the partnership's income, expenses, gains and losses**, each partner **taxed according to their respective share**
- Hedge funds aim to **consistently achieve positive returns under all market conditions**, and **do not necessarily hedge all risk as the name might imply**

Investment manager

Determines strategy, makes investing decisions and allocations, and manages portfolio risk. The manager is also **invested in the fund** and is compensated via a **management fee plus a performance fee** based on the fund's annual performance. **Managers only get a performance fee if the fund makes money above a certain benchmark.**

Prime broker

Funds **must secure their loans with collateral** to gain margin and execute trades. Each broker — usually a **large securities firm** — uses **its own risk matrix** to determine how much to lend each client. **Brokers are themselves subject to regulations, which act as indirect regulations on the hedge funds doing business with them.**

FUNDS OF HEDGE FUNDS Funds that **hold a portfolio of hedge funds**, creating a diversified portfolio accessible to **smaller investors** or to those **without the resources, time or expertise** to choose among managers. They are assumed to have **expertise in conducting due diligence** and may be able to **negotiate better redemption terms** than individual investors. They may diversify across **strategies, investment regions and management styles**. **The distinction between a single hedge fund and a fund of funds is not necessarily clear-cut — hedge funds can invest in other hedge funds.**

§2.2.2 · HEDGE FUND STRATEGIES — FOUR GROUPS, TWENTY-PLUS STRATEGIES

Hedge fund strategies arise from taking **speculative positions in a combination of market and credit risk instruments** for which the manager believes the **risk-reward relationship is attractive**. Such positions are often referred to as **arbitrage, or risk arbitrage**.

GROUP	OBJECTIVE AND SUBCATEGORIES
Equity hedge	Seek long-term capital appreciation while maintaining low net exposure to the overall stock market or individual industry groups; profit from long or short positions in publicly traded equities and equity derivatives. Market neutral — technical or fundamental analysis to hold undervalued equities long and overvalued equities short in approximately equal amounts, profiting from relative price movements without exposure to market risk. Fundamental growth — buy high-growth companies expected to sustain high capital appreciation. Fundamental value — buy shares believed undervalued; it is the hedge fund structure, not the assets, that makes this an alternative investment. Quantitative directional — long undervalued, short overvalued based on technical analysis; market exposure varies with the relative size of the two books. Short bias — predominantly short overvalued equities, possibly with smaller longs, but negative market exposure overall.
Event driven	Concentrate on the profit potential created by major corporate events — mergers, acquisitions, restructurings, bankruptcies or liquidations — through long or short positions in common equity, preferred equity or debt. Merger arbitrage — buy the shares of the firm being acquired and sell short the acquirer. Distressed / restructuring — buy the undervalued securities of firms in financial distress where analysis shows a successful restructuring will add value; possibly short overvalued security types simultaneously. Activist shareholder — buy enough equity to influence company policy and increase company value. Special situations — firms issuing or repurchasing securities, spinning off divisions, selling assets or distributing capital.
Relative value	Profit from a pricing discrepancy — an unusual short-term relationship — between related securities, expecting the discrepancy to resolve over time. Fixed-income convertible arbitrage — market-neutral (a theoretical zero-beta portfolio); exploit mispricing between a convertible bond and its component parts by buying the convertible debt and simultaneously selling the same issuer's common stock. Fixed-income asset backed — relative value across ABS and MBS. Fixed-income general — long/short between two corporate issuers, corporate versus government, different parts of one capital structure, or different parts of a yield curve. Volatility — use options to go long or short market volatility, with dynamic hedging to offset related risks. Multi-strategy — relative value within and across asset classes, looking for opportunities wherever they exist.
Macro	Based on global economic trends and events; long or short positions in equities, fixed income, currencies or commodities. Emphasise a top-down approach to identify economic trends evolving across the world. Trade the most liquid markets — currencies and government bonds — betting on macroeconomic events such as changes in interest rate policies or currency devaluations, relying mostly on an assessment of economic fundamentals.

§2.2.3 · HEDGE FUND FEES — THE WORKED EXAMPLE

Fee levels

The common structure was once “**2 and 20**” — a **2% management fee** and a **20% incentive fee**, both paid by LP investors. The **average industry fee is now closer to 1.6% and 17.75%**. **Funds of hedge funds** may charge a further **1% management fee and 10% incentive fee** — “**1 and 10**” — **in addition to** the fees charged by the individual hedge funds inside the structure.

Profit definitions

Profits can be defined as **(1) any gains in value, (2) any gains in value in excess of the management fee, or (3) gains in excess of a hurdle rate**. The incentive fee may be calculated **on profits net of management fees** or **independent of them**. The question will say which.

ABC Funds — guide §2.2.3

ABC Funds has a value of **R110 million at initiation**. It charges a **2% management fee** based on AUM **at the beginning of the year** and a **20% incentive fee with a 5% hurdle rate**, and **uses a high-water mark**. Incentive fees are calculated on **gains net of management fees**. Ending values **before fees**: **Year 1 R102.2m, Year 2 R118.0m**.

Year 1

Management fee — $R110.0m \times 2\%$	R2.2m
Return net of management fee — $(R102.2m - R2.2m) \div R110.0m - 1$	-9.1%
Incentive fee — the return is less than the hurdle rate	R0
Total fees	R2.2m
Ending value net of fees — $R102.2m - R2.2m$	R100.0m

Year 2

Management fee — $R100.0m \times 2\%$	R2.0m
Return net of management fee — $(R118.0m - R2.0m) \div R100.0m - 1$	+16.0%
Incentive fee — $(R118.0m - R2.0m - \mathbf{R110.0m}) \times 20\%$	R1.2m
Total fees — $R2.0m + R1.2m$	R3.2m
Net return — $(R118.0m - R3.2m) \div R100.0m - 1$	+14.8%

EXAM WATCH — WHERE THE MARKS ARE LOST In Year 2 the incentive fee is **calculated based on gains in value above R110 million, because that is the high-water mark** — **not** above the R100 million the year opened with. Substituting R100m there is the single most common error in this calculation.

Potential benefits and risks

Hedge fund returns have tended to be **better than global equities in down markets** and to **lag global equities in up markets**. **Different strategies have the best returns during different time periods**. Statements about performance and diversification benefits are **problematic because of the great variety of strategies used**. Less-than-perfect correlation with global equity returns may offer some diversification benefit, **but correlations tend to increase during periods of financial crisis**.

Institutional investors' **number one objective** is returns **uncorrelated to equity**. Correlation with traditional stocks runs **20–35%** for futures, fixed arbitrage and global macro funds, up to **70–80%** for long/short or emerging equity funds.

Valuation

Values are based on **market values for traded securities** but must use **model (estimated) values for non-traded securities**. It is **most conservative to use the prices at which a position could be closed — bid prices for long positions and ask prices for short positions**; some funds use **the average of bid and ask** instead. For **illiquid securities**, quoted prices may be **reduced for the degree of illiquidity**, based on position size relative to total value outstanding and average trading volume.

Due diligence — what to examine

Investment strategy · investment process · source of competitive advantages · historical returns · valuation and returns calculation methods · longevity · amount of assets under management · management style · key person risk · reputation · growth plans · systems for risk management · appropriateness of benchmarks.

The analysis is challenging: a **lack of persistence in returns** means past outperformers may not outperform in future, and reputation, risk systems and management style are **difficult to quantify**. Previously profitable strategies **become less profitable as more funds pursue them**. One further factor: the **percentage of total AUM held by any single investor — a large withdrawal may cause the fund to fail**, especially if assets are illiquid. Transparency is limited: funds treat strategies as **proprietary** and **regulatory disclosure requirements are minimal**.

Hedge funds in South Africa

Internationally hedge funds have been around **about 60 years**; in SA they are a **relatively newer phenomenon**. Following the development of **securities lending in the early 1990s**, some hedge trading developed in proprietary funds. **By 2000** there were approximately **20–30 hedge funds (R2.5bn)**. Growth was slow — **120 funds in 2005 with R10bn AUM**, to **140 managers and R60bn+ AUM in 2016**. In 2017 assets moved to **R62bn** according to the Novare survey.

Drivers for growth: regulatory development — Regulation 28, covering pension fund investment into hedge funds, and regulation of hedge funds under **CISCA** as either a Retail Investor Hedge Fund or a Qualified Investor Hedge Fund — and **demand from international investors** for diversified investment exposure.

EXAM WATCH · THE CISCA CLASSES **Retail Investor Hedge Funds (RHF)** — minimum **R50,000**; **priced daily; withdraw within a month; more prescriptive** regulation. **Qualified Investor Hedge Funds (QIHF)** — experienced or institutional investors with **R1 million or more**; **priced monthly; one calendar month's notice**; managers have **greater freedom**.

§2.3 · PRIVATE EQUITY — STRATEGIES AND LEVERAGED BUYOUTS

Private equity — investment in **privately owned companies**, or in **public companies with the intent to take them private**. The focus of private equity firms, which may manage many funds, **changes over time as business conditions and the availability of financing change**.

Leveraged buyouts

Venture capital

Development capital

Distressed investing

The majority of private equity funds invest either in **private companies or public companies they intend to take private (LBO funds)**, or in **early stage companies (venture capital funds)**. Two additional, smaller categories are **distressed investment funds** and **developmental capital funds**. A private equity fund may **also charge fees for arranging buyouts, fees for a deal that does not happen, and fees for handling asset divestitures after a buyout**.

Leveraged buyouts — the most common type

“Leveraged” refers to the fact that the fund's purchase of the portfolio company is funded primarily by debt. This may be:

- **Bank debt** — leveraged debt
- **High-yield bonds**
- **Mezzanine financing** — **debt or preferred shares that are subordinate to the high-yield bonds issued** and that **carry warrants or conversion features** giving investors participation in equity value increases

MBO — management buyout

The **existing management team** is involved in the purchase.

MBI — management buy-in

An **external management team** will replace the existing management team.

How value is created — and what makes a good candidate

In an LBO the private equity firm seeks to increase firm value through some combination of **new management, management incentives, restructuring, cost reduction or revenue enhancement**. **Firms with high cash flow are attractive LBO candidates, because their cash flow can be used to service and eventually pay down the debt taken on for the acquisition.**

EXAM WATCH Buyout investors take a majority control in acquired companies; venture capitalists only take a minority interest. That one line separates the two strategies in most multiple-choice framings.

§2.3 · VENTURE CAPITAL STAGES AND EXIT STRATEGIES

Venture capital funds invest in companies **in the early stages of their development**. The investment is often **equity but can be convertible preferred shares or convertible debt**. While the risk of start-ups is often great, **returns on successful companies can be very high** — often when a company is grown to the point of being sold, at least in part, to the public via an **IPO**. Investee companies are the fund's **portfolio companies**, and **VC managers are closely involved in their development, often sitting on boards or filling key management roles**.

STAGE	WHAT THE MONEY DOES
Formative — angel investing	Investments made very early in a firm's life, often the “idea” stage . Funds are used for business plans and assessing market potential . The funding source is usually individuals — “angels” — rather than venture capital funds .
Formative — seed stage	Investments for product development, marketing and market research . This is typically the stage at which venture capital funds make their initial investments , through ordinary or convertible preferred shares.
Formative — early stage	Investments made to fund initial commercial production and sales .
Later stage	The company already has production and sales and is operating as a commercial entity . Funds are typically used for expansion of production and/or increasing sales through an expanded marketing campaign .
Mezzanine-stage financing also called bridge	Capital provided to prepare the firm for an IPO — the bridge between the expanding company and the IPO. The term refers to the timing of the financing — between private company and public company — rather than the type of financing .

What venture capital looks for

Exponential growth potential · an attractive industry · a sustainable advantage platform · an excellent team with “execution” · owners receptive to the involvement of outsiders · owners willing to share the wealth creation · credible exit alternatives four to seven years out.

Exit strategies — the average holding period is five years

EXIT	MECHANISM
Trade sale	Sell a portfolio company to a competitor or another strategic buyer .
IPO	Sell all or some shares of a portfolio company to the public .
Recapitalisation	The company issues debt to fund a dividend distribution to equity holders — the fund. This is not an exit — the fund still controls the company — but is often a step toward one .
Secondary sale	Sell a portfolio company to another private equity firm or a group of investors .
Write-off / liquidation	Reassess and adjust to take losses from an unsuccessful outcome.

§2.4 · REAL ESTATE — THE ASSET TYPES

Investment in real estate can provide **income in the form of rents** as well as the potential for **capital gains**. As an asset class it can provide **diversification benefits** and a **potential inflation hedge**, because rents and values tend to increase with inflation. The three asset types: **residential property** (single-family homes), **commercial property** (produces income), and **mortgages** (loans with residential or commercial property as collateral).

Residential property

Considered a **direct investment** in real estate. Some buyers pay cash but most take a **mortgage**. The **issuer (lender) of the mortgage has a direct investment in a whole loan and is said to “hold the mortgage”**. Issuers often sell the mortgages they originate; those are then **pooled (securitised) as publicly traded mortgage-backed securities (MBS)**, which represent an **indirect investment in the mortgage loan pool**. Property purchased with a mortgage is a **leveraged investment**, and **the owner's equity is the property value minus the outstanding loan amount** — so changes in property value over time **affect the owner's equity**.

Commercial real estate

Generates **income from rents**. **Homes purchased for rental income are considered investment in commercial property**. Large properties such as an office building are a **direct investment for institutions or wealthy individuals**, purchased for cash or leveraged with a mortgage. **Long time horizons, illiquidity, the large size of investment needed and the complexity of the investments make commercial real estate inappropriate for many investors**. It can also be held by a **limited partnership** — partners have limited liability and the general partner manages the investment and the properties — or by a **REIT**.

REIT — real estate investment trust — issues **shares that trade publicly like shares of stock**.

REITs are often identified by the **type of real estate assets they hold**: mortgages, hotel properties, malls, office buildings or other commercial property. **Income is used to pay dividends**, and **typically 90% of income must be distributed to shareholders to avoid taxes** that the REIT would otherwise pay before distribution.

WHY REITS EXIST The REIT market **has expanded to overcome many of the problems of direct investment**. Shares are **actively traded and are more likely to reflect market value**; investing in a REIT gives **exposure to a diversified real estate portfolio**; and **investors do not need property management expertise, because the REIT manages the properties**.

Eight characteristics that complicate measurement

CHARACTERISTIC	WHY IT MATTERS
Heterogeneity	Bonds from one issue are alike, as are the shares of one company — but no two properties are exactly the same , because of location, size, age, construction materials, tenants and lease terms.
High unit value	Real estate is indivisible , so unit value is significantly higher than for stocks and bonds, making a diversified portfolio difficult to construct .
Active management	Private real estate investment requires active property management — maintenance, negotiating leases, collecting rents. Management costs must be considered either way.
High transaction costs	Buying and selling involves appraisers, lawyers, brokers and construction personnel .
Depreciation and desirability	Buildings wear out over time and may become less desirable because of location, design or obsolescence .
Cost and availability of debt capital	Values are impacted by the level of interest rates and the availability of debt capital — usually lower when rates are high and debt capital is scarce .
Lack of liquidity	It takes time to market and complete the sale of a property.
Difficulty in determining price	Heterogeneity and low transaction volume mean appraisals are usually necessary — and even then values are often based on similar, not identical, properties . Limited market participants and lack of local knowledge make it hard for an outsider to value property, so the market is less efficient — which means investors with superior information and skill may have an advantage in exploiting the inefficiencies .

THE PAY-OFF FOR THAT INEFFICIENCY Because appraisals rest on **similar rather than identical** properties and few participants know a local market well, real estate is **less efficiently priced than listed securities** — which is exactly why **investors with superior information and skill may have an advantage in exploiting the market inefficiencies**.

Five reasons to invest

- **Current income** — rents, after operating expenses, financing costs and taxes
- **Capital appreciation** — property values expected to rise over time
- **Inflation hedge** — both rents and values expected to rise with inflation
- **Diversification** — less than perfectly correlated with stocks and bonds
- **Tax benefits** — in some countries the **depreciable life is shorter than the actual life**, so depreciation expense is higher and taxable income lower; and **REITs do not pay taxes in some countries**, letting investors **escape double taxation**

The principal risks

Business conditions — GDP, employment, household income, interest rates, inflation · **new property lead time** — conditions can weaken between approval, completion and full lease-up · **cost and availability of capital** · **unexpected inflation** — some leases pass it through, but values may not keep up when markets are weak · **demographic factors** · **lack of liquidity** — a quick sale requires a **significant discount** · **environmental issues** — contamination by a prior or adjacent owner · **availability of information** · **management expertise** · **leverage** — measured by the **loan-to-value (LTV) ratio**; higher LTV means higher risk since **lenders have a superior claim on default**, and a **small decrease in NOI is magnified** in the cash flow left to equity · **other** — defects, natural disasters, terrorism.

Identified risks can be **hedged using insurance** or **shifted to tenants** — a lease could require the tenant to **reimburse unexpected operating expenses**.

EXAM WATCH Two real estate traps recur. First, **homes purchased for rental income are an investment in commercial property**, not residential — the classification follows **the income, not the building type**. Second, on **leverage**: a higher LTV means higher risk **because lenders hold a superior claim in the event of default**, and because a **small fall in net operating income is magnified** in whatever cash flow reaches the equity holder after debt service.

Direct investment vs the REIT — a summary

	DIRECT / PRIVATE	REIT
Liquidity	Illiquid — slow to market and sell	Actively traded like a share
Pricing	Appraisal-based, from similar properties	More likely to reflect market value
Diversification	Hard — high indivisible unit value	Exposure to a diversified portfolio
Management	Requires active management by the owner or an agent	The REIT manages the properties
Tax	Depreciation shelters income in some countries	90% of income must be distributed

\$2.5 · COMMODITIES — ROUTES TO EXPOSURE

Commodities are **real assets, mostly natural resources** — agricultural products, oil, natural gas, precious and industrial metals. They are considered a **hedge against inflation** as they are **not sensitive to public equity markets**, and their value rises and falls with **supply and demand**. From politics and weather to disease and speculation, commodities are **among the most volatile instruments in the financial world**.

WHY DERIVATIVES DOMINATE While it is possible to invest directly in commodities such as grain and gold, **the most commonly used instruments to gain exposure to commodity prices are derivatives** — because commodities are physical goods and therefore **incur costs for storage and transportation**. **Returns are based on price changes and not on income streams.**

ROUTE	NOTE
Direct physical investment	Unless the investment comprises commodities such as Krugerrand gold coins , investors would require warehouses to store the commodities — generally considered impractical and uneconomic . Physical investment is also limiting for investors seeking broad exposure .
Shares of commodity producers	Exposes the investor not only to commodities but to the financial structure, governance and management skills and practices of the company . In addition the company may validly hedge its own production , so investors will not get the full benefit of changes in commodity prices .
Commodity futures	Purchase futures contracts to simulate ownership . By periodically rolling over the contracts prior to expiry and reinvesting in new futures, investors obtain returns equal to the return from a single commodity or an index of several .
Commodity index funds	Let investors buy a balanced and diversified basket of commodities in a single investment . Index funds do not have physical ownership — they use futures to buy a forward position, sell it as it approaches expiry, and use the proceeds to buy forward again by one or two months . The most liquid index is the Standard & Poor's Goldman Sachs Commodities Index (S&P GSCI) .
ETFs and ETNs	Exchange traded funds (ETFs) are traded instruments representing ownership in an underlying portfolio of commodities . Exchange traded notes (ETNs) are traded as debt securities — a promise to pay the investor an amount linked to a given benchmark , so the investor takes on the credit risk of the issuer . There are currently no commodity-based ETFs or ETNs in South Africa .
Exchange traded commodities (ETCs)	Vehicles that track the performance of a single underlying commodity — gold, oil, zinc — or a commodity index . Traded and settled exactly like normal shares , with dedicated ETC sectors on the London Stock Exchange, Euronext Amsterdam and Deutsche Börse . Generally benchmark a total-return index . The first ETC was a Gold ETC listed in 2003 on the Australian Securities Exchange; an ETC was first listed on the LSE in 2004 .

The six sectors

Energy — crude oil, natural gas and refined products; **the sector with the greatest market value**. **Industrial metals** — aluminium, nickel, zinc, lead, tin, iron and copper, plus **rare-earth metals** used increasingly in electronics. **Grains** — wheat, corn, soybeans, rice. **Livestock** — pigs, sheep, cattle, poultry. **Precious metals** — gold, silver, platinum. **Softs (cash crops)** — coffee, sugar, cocoa, cotton.

The three types of investor

Institutional — insurers, banks, pension funds, investment companies, collective investment schemes, hedge funds and endowments; **most hold commodities as a non-essential diversifying investment**. **Private** — high net worth and sophisticated individuals, for an **inflation hedge and diversification**. **Retail** — **limited direct exposure**, commodities traditionally regarded as **too volatile**; internationally **indirect exposure comes through the assets institutions hold on their behalf**, such as pension funds.

Two sector dynamics worth knowing

Crude oil can be **stored indefinitely by keeping it in the ground**, and in tanks and tanker ships; many countries hold **strategic reserves**. The two most traded grades are **Brent Crude and West Texas Intermediate (WTI)**. Supply is affected by **drilling and extraction technology, economic cycles, alternative energy efficiency, environmental restrictions** and **political risk** — **over half the crude supply comes from the Middle East**.

Industrial metals demand is **primarily tied to GDP growth and business cycles**, through construction and manufacturing.

Storage of metals is not costly. **Union strikes and environmental regulation** can significantly affect supply; mines and smelters are **large-scale operations with high development and fixed costs**.

§2.5.4–2.5.5 · COMMODITY PRICES, VALUATION AND THE SOURCES OF RETURN

Spot prices for commodities are a function of supply and demand. Demand is affected by the **value of the commodity to end-users** and by **global economic conditions and cycles**. Supply is affected by **production and storage costs and existing inventories**. **Both supply and demand are affected by the purchases and sales of non-hedging investors — speculators.**

For many commodities **supply is inelastic in the short run**, because of the **long lead times to alter production levels** — drilling oil wells, planting crops, or deciding to plant less of them. As a result **commodity prices can be volatile when demand changes significantly over the economic cycle**. Production of some commodities, **especially agricultural**, can be **significantly affected by the weather** — high prices when production is low, low prices when production is high. Costs of extracting oil and minerals **increase as more expensive methods or more remote areas are used**.

What each side analyses

Producers analyse **economic events, government policy and forecasts of future supply** to estimate future needs. **Investors** analyse **inventory levels, forecasts of production, changes in government policy and expectations of economic growth** in order to forecast commodity prices.

WHY THE FORMULA HAS THE SHAPE IT DOES Wheat today and wheat six months from today are **different products**. Buying the commodity today gives the buyer **the use of it if needed**; contracting for delivery in six months **avoids storage costs and having cash tied up**. The cost-of-carry equation prices exactly that difference.

Valuation — the cost-of-carry equation

$$\text{futures price} = \text{spot price} (1 + \text{risk-free rate}) + \text{storage costs} - \text{convenience yield}$$

Contango vs backwardation

Convenience yield is the **value of having the physical commodity for use** over the period of the futures contract. **If this equation does not hold, an arbitrage transaction is possible.**

Little or no convenience yield → futures higher than spot → **contango**.

High convenience yield → futures less than spot → **backwardation**.

Three sources of futures returns

Roll yield — from the difference between spot and futures, or between two futures of different expiry, as futures converge toward spot near expiration. **Positive for a market in backwardation, negative for a market in contango.**

Collateral yield — the **interest earned on collateral** required to enter the futures contract. **Change in spot prices** — total price return combines the **spot move and the convergence of futures to spot**.

BENEFITS AND RISKS Returns have been **lower than global stocks or bonds** and **Sharpe ratios low**. But correlations with global equities and bonds are **historically low** — **typically less than 0.2**, so commodities **diversify**; and because prices **move with inflation** they **hedge inflation risk**. To the extent they track inflation the **real return would be zero, though futures may offer positive real returns**. **Supply is inelastic in the short run** — long lead times to drill or plant — which is why prices are volatile over the cycle.

Blockchain — a **time-stamped series of an immutable record of data managed by a cluster of computers that are not owned by a central authority**. A blockchain is **transparent**, and **everyone involved is accountable for their actions**. **Blockchain is the technology that enables the existence of cryptocurrency**, among other things.

Cryptocurrency — a **digital or virtual currency secured by cryptography**; cryptocurrencies can also be thought of as **decentralised networks based on blockchain technology**. A cryptocurrency is a **medium of exchange**, such as the US dollar, **but digital**, using encryption techniques to **control the creation of monetary units and to verify the transfer of funds**.

Why it became an alternative investment

Cryptocurrency has become an increasingly popular alternative investment **thanks to its relative accessibility**. Because it is based on blockchain networks used for security, it is **almost impossible to double-spend cryptocurrency or for it to be counterfeit**. It also **operates entirely independently from the government, making it unaffected by interference**.

The drawbacks: that same independence **has in some cases led to its use in illegal activities**, and while it is thought to be extremely secure, **the networks are still subject to some vulnerabilities**.

There are **over 800 different cryptocurrencies** currently on the market according to **Coin Market Cap**, a platform where cryptocurrencies are **ranked based on their total market volume**. **The most well-known example is Bitcoin, which is also the most valuable today**.

Bitcoin defined

- The **first decentralised digital currency and worldwide payment system**, created by a **slow computer process called mining**
- The international currency is **not regulated by banks** and can be used to **purchase goods from anywhere in the world**
- **There is only 21 million Bitcoin that can be mined in total**, and **this creates demand as adoption increases**
- It is **decentralised digital cash for the internet, not controlled by a central government or federal reserve, but by the people who invest in it**
- **All transactions are processed by miners**, creating a **digital ledger of secure and mostly anonymous transactions between individuals**

§2.6.3–2.6.4 · BENEFITS OF CRYPTOCURRENCY AND FUTURE SCENARIOS

BENEFIT	MECHANISM
Fraud	Cryptocurrencies are digital and cannot be counterfeited or reversed arbitrarily by the sender , as with credit card chargebacks .
Identity theft	When you give a credit card to a merchant you give them access to your full credit line , even for a small transaction — credit cards operate on a “ pull ” basis, where the store initiates the payment and pulls the designated amount from your account. Cryptocurrency uses a push mechanism that allows the holder to send exactly what they want to the recipient with no further information .
Immediate settlement	Purchasing real property typically involves third parties, delays and fees . In many ways the blockchain is a “ large property right database ”. Bitcoin contracts can be designed and enforced to eliminate or add third-party approvals, reference external facts, or be completed at a future date or time — for a fraction of the expense and time required for traditional assets .
Access to everyone	There are approximately 2.2 billion individuals with access to the internet or mobile phones who do not currently have access to traditional exchange systems . These individuals are primed for the cryptocurrency market .
Lower fees	There are usually no transaction fees for the cryptocurrency exchange, because the network compensates the miners . Even so, most users go through an online exchange, and those are likely to charge fees .

Future scenarios the guide names

Bitcoin, Ethereum, Ripple, Litecoin and others continue to experience **massive growth in price, market capitalisation and mainstream adoption**. It is **no longer a question of whether cryptocurrencies are disrupting the global economy, but by how much and what the future holds**.

The US Federal Reserve could issue its own digital currency, as some global central banks are exploring. **Large companies such as Amazon, Walmart and Starbucks might issue digital coins** that inspire trust and gain wide acceptance. **Retail giants, by accepting payments in the currency, could elevate Bitcoin, Ethereum or another cryptocurrency above the others** vying to offer safety, soundness and utility. And finally, **if trust is lost in government-backed, or fiat, currencies, a cryptocurrency future could come about by default**.

THE LIQUIDITY PREMIUM The deck's framing for the whole of Chapter 2: alternative investments are **less liquid** than traditional ones, and the extra return investors demand for accepting that illiquidity is the **liquidity premium**. It is the single idea that ties together hedge fund lockups and notice periods, private equity's five-year holding period, and real estate's slow sale at a discount.

Option price sensitivities

IF THIS RISES...	CALL	PUT
Price of the underlying	↑	↓
Exercise / strike price	↓	↑
Risk-free rate	↑	↓
Volatility	↑	↑
Time to expiration	↑	↑
Benefits of holding the asset (dividends, coupons)	↓	↑

The formulas

QUANTITY	FORMULA
Futures price	$F = S (1 + r)^t$
Intrinsic value of a call	Market price – Strike price
Intrinsic value of a put	Strike price – Market price
Call payoff	$\max(0, S_T - X)$
Put payoff	$\max(0, X - S_T)$
Profit on an option	Payoff – Premium
Breakeven on a long call	Strike + Premium
Commodity futures price	Spot (1 + risk-free rate) + storage costs – convenience yield
Hedge fund management fee	Beginning-of-year AUM × fee %
Hedge fund incentive fee	(Gross ending value – management fee – high-water mark) × incentive %

IF YOU ONLY REVISE ONE THING Forward commitment = obligation, symmetrical payoff, both parties bound. Contingent claim = right, asymmetrical payoff, and the buyer's loss is capped at the premium. Nearly every derivatives question in this module is a variation on that one distinction.

APPENDIX · THE TRAPS

TRAP	THE RIGHT ANSWER
"Hedge funds hedge risk"	They do not necessarily hedge risk as the name might imply — they seek positive returns under all market conditions.
Short selling vs going short	Short selling = the spot/cash market. Going short = the derivatives market.
Settlement date	Normally two days before the final expiration date — not the same day.
Strike vs spot	The price specified in a derivative contract is not the current or spot price — it is a price good for future delivery .
American vs European	American = on or before expiry, and is the most common type . European = only on the expiration date.
Swap principal	The principal usually does not change hands during or at the end of the swap — only the cash flows are exchanged.
Netting	Interest rate swap payments are netted . Currency swap payments are not — they are in different currencies .
Payer vs receiver swaption	Payer = pay fixed / receive floating; buyer hopes the fixed rate goes up . Receiver = receive fixed / pay floating; buyer hopes it goes down .
Roll yield sign	Positive for a market in backwardation, negative for a market in contango.
Mezzanine-stage financing	Refers to the timing — between private and public company — not the type of financing.
Recapitalisation	Not an exit — the fund still controls the company. It is often a step toward one.
High-water mark base	The incentive fee is computed on gains above the high-water mark , not above the value the year opened with.
Homes bought to rent	Count as an investment in commercial property, not residential.
ETN credit risk	An ETN is a debt security — the investor takes on the issuer's credit risk . An ETF, representing ownership of a portfolio, does not.
Angel funding source	Angel investing is funded by individuals , not by venture capital funds — VC funds typically come in at the seed stage .
JSE vs BESA regulation	The JSE regulates exchange trading in derivatives; the BESA does not — it provides the platform, regulating fixed-income derivatives through Intersec .

SOURCES FMP V05 Mod06 Derivatives & Alt Investments STUDY GUIDE 25 10 06 CJ (60pp) and FMP V06 Mod06 Part Derivatives & Alt Inv CLASS PRES 26 09 02 CJ (122 slides). The **guide is the primary MCQ source**; the deck carries the **futures pricing formula, the breakeven arithmetic and the CISCA RHF/QIHF split** the guide omits. Section numbers are the guide's own.